

ALEXANDRE MANUEL RIBAS SOUSA

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PROFESSIONAL SUMMARY

An engineer and researcher focused on the intersection of system integration, IoT, electronics, computer vision and Applied AI. Currently serving as a Research Assistant in Computer Vision & AI at UTAD University, I work across Edge systems, image annotation, data acquisition pipelines and preliminary object detection model training. My research approach combines critical thinking, technical autonomy and hands-on development across AI, cybersecurity and DevOps, with practical experience in Docker, Kubernetes and AI frameworks. I am motivated by scientific development and by translating emerging technologies into reliable solutions.

WORK EXPERIENCE

Research Assistant (Computer Vision & AI) – UTAD University *04/2026 – Present | Vila Real, Portugal*

- Designed and prototyped Edge systems for computer vision and AI applications under practical hardware and processing constraints.
- Annotated and curated image datasets to support computer vision workflows and the development of object detection models.
- Built data acquisition pipelines to collect, organize and prepare reliable datasets for subsequent model development and evaluation.
- Performed preliminary training and experimentation with one-stage object detection models for computer vision applications.
- Contributed to scientific research and development through technical implementation, experimentation and documentation of results.

Research Assistant (Applied AI & Data Engineering) – INESC TEC *07/2025 – 01/2026 | Porto, Portugal*

- Leveraging data engineering principles to optimize hospital triage systems.
- Development and training of predictive models using Machine Learning to predict critical outcomes for clinical decision support.
- Engineered robust ETL pipelines to process noisy clinical datasets, ensuring data quality for downstream Machine Learning models.
- Implemented Applied AI solutions using Large Language Model (LLM) frameworks (LangChain) for symptom analysis.
- Collaborated with cross-functional teams (medical & technical), translating complex requirements into technical/practical implementations.

Invited Assistant Professor (Electronics & Sensors) – UTAD University *05/2024 – Present | Vila Real, Portugal*

- Lecturing undergraduate and master's degree classes such as "Instrumentation and Sensors" and "Power Electronics", covering the physical principles of data acquisition, signal conditioning and AC-DC converters.
- Co-supervision of students in the "Duckietown" robotics project, guiding them through hardware-software integration challenges and debugging sessions.
- Development of practical laboratory teaching material in Power Electronics.
- Implemented dynamic learning approaches, improving student engagement and problem-solving skills.

Research Assistant (IoT & Secure Architectures) – UTAD (A-MoVeR Project) *03/2023 – 07/2024 | Vila Real, Portugal*

- Autonomously designed and deployed a secure infrastructure for Smart Mobility.
- Self-taught and implemented a container-native Identity Access Management (IAM) framework using Keycloak and Kubernetes to secure heterogeneous IoT devices.
- Authored scientific papers validating the architecture, culminating in a 19/20 Master's Thesis for delivering a fully functional, scalable prototype.

DevOps Engineer Intern (Infrastructure Automation) – Swiftec *05/2023 – 02/2024 | Vila Real, Portugal*

- Focus on efficiency and eliminating manual redundant tasks.
- Automated manual server deployments using Terraform and Ansible, significantly reducing deployment time and human error.
- Set up monitoring dashboards (Grafana/Prometheus) to provide real-time visibility into system health, allowing for proactive maintenance.
- Demonstrated adaptability by quickly mastering internal tools and Linux environments to support the senior engineering team.

EDUCATION

University of Trás-os-Montes and Alto Douro (UTAD) – Vila Real/Portugal *2018 – 2024*

Integrated Master's in Electrical and Computer Engineering

Final Grade: 19/20 (Dissertation). Title: “Development of Secure Systems for IoT in Green Mobility Environments: An Approach to Authentication, Authorization and Auditing”

VOLUNTEERING & LEADERSHIP

Volunteer – Casa de Saúde do Bom Jesus *January 2018 – December 2018 | Braga/Portugal*

- Assisted in the daily care and social integration of patients, fostering a supportive environment through organized recreational activities and direct emotional support.

Director – NEUTAD *Vila Real*

- Organizational Leadership: Directed the strategic and operational management of the Student Association, acting as the primary liaison between the student body, faculty, and the business community to advocate for educational interests.
- Event Management & Skill Development: Spearheaded the planning and execution of technical workshops (e.g., IoT, AutoCAD, LTspice) and lecture series. Successfully engaged over 50 students, bridging the gap between theoretical academic curriculum and market-demanded practical skills.
- Resource & Inventory Administration: Managed the technical inventory of electronics and laboratory kits, ensuring logistical support for student-led projects while organizing social events to foster team spirit and networking.

CERTIFICATIONS

Machine Learning & Deep Learning: (NumPy, Pandas, DL...) – IA Expert Academy
LTspice, AutoCAD – NEUTAD-UTAD
Entrepreneurship – SPI (Portuguese Innovation Society)

SKILLS

Computer Vision & Edge AI: Edge Systems, Image Annotation, One-Stage Object Detection, YOLO
Applied AI & Data Ops: Inference Pipelines, MLOps basics (deploying models), LangChain, n8n, Data Cleaning/ETL (Pandas), YOLO
IoT & System Integration: Sensor Interfacing (I2C, SPI, UART), Protocol Implementation (MQTT, LoRaWAN), Schematic Reading & Prototyping, Hardware-Software debugging
Infrastructure & Automation: Containerization (Docker), Orchestration (Kubernetes - K8s), Infrastructure as Code (Terraform, Ansible), Linux System Administration
Tools & Languages: Python (Advanced), C/C++ (Intermediate), Bash Scripting, LTSpice, Git, MATLAB, Java

KEY PUBLICATIONS

“Edge-Based AI for Real-Time Threat Detection in 5G-IoT Networks: A Comparative and Architectural Review”
- https://ieeexplore.ieee.org/document/11140446

“A Container-Native IAM Framework for Secure Green Mobility: A Case Study with Keycloak and Kubernetes”
- https://www.mdpi.com/2078-2489/16/9/802?utm_source=researchgate.net&utm_medium=article

"5G Security Features, Vulnerabilities, Threats, and Data Protection in IoT and Mobile Devices: A Systematic Review"
- https://link.springer.com/chapter/10.1007/978-3-031-76459-2_20

LANGUAGES

Portuguese (Native), English (C1 - Fluent)